UNIVERSAL OPENNESS OF SPEC(B) OVER SPEC(A) TESTED ON ONE EXPLICIT PRINCIPAL OPEN FOR A FINITE GENERATING SET

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Editorial note. Statement span is the whole lemma environment lines 21846-21852; proof span is lines 21854-21875, the immediately following proof environment.

Editorial note. The proof contains two xymatrix displays; the retained preamble provides the Xy-pic setup, so no extra_packages are declared.

Editorial note. No external_references are needed: the proof cites no label and no bibliography entry.

Editorial note. The proof text is retained verbatim, including the mild imprecision in the original (the statement writes x_1,...,x_d while the proof writes x_1,...,x_n); no correction was applied.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Base change of morphisms of schemes, the fact that principal opens D(g) form a basis of the topology, polynomial algebras and the cartesian square Spec(B[x_1..x_n]) / Spec(A[x_1..x_n]) against Spec(B') / Spec(A'). All are standard; the reduction itself is retained.

Lemma 0.1. *Let $A \to B$ be a ring map. Say B is generated as an A-module by $b_1, \ldots, b_d \in B$. Set $h = \sum x_i b_i \in B[x_1, \ldots, x_d]$. Then* $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *is universally open if and only if the image of $D(h)$ in* $\mathrm{Spec}(A[x_1, \ldots, x_d])$ *is open.*

Proof. If $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is universally open, then of course the image of $D(h)$ is open. Conversely, assume the image U of $D(h)$ is open. Let $A \to A'$ be a ring map. It suffices to show that the image of any principal open $D(g) \subset \mathrm{Spec}(A' \otimes_A B)$ in $\mathrm{Spec}(A')$ is open. We may write $g = \sum_{i=1,\ldots,d} a'_i \otimes b_i$ for some $a'_i \in A'$. Let $A[x_1, \ldots, x_n] \to A'$ be the A-algebra homomorphism sending x_i to a'_i. This also induces a B-algebra homomorphism $B[x_1, \ldots, x_n] \to A' \otimes_A B$ sending h to g. Since

$$\begin{array}{ccc} \mathrm{Spec}(B[x_1, \ldots, x_n]) & \longleftarrow & \mathrm{Spec}(B') \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A[x_1, \ldots, x_n]) & \longleftarrow & \mathrm{Spec}(A') \end{array}$$

is cartesian the image of $D(g)$ in $\mathrm{Spec}(A')$ is equal to the inverse image of U in $\mathrm{Spec}(A')$ and hence open. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.